LESSON PLAN

A/C ENGINEERING SYSTEM (17)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	25
Total No. of Lesson Plans	30	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Types of fire zone classification of a/c, detection system requirements, types of detection system, types of detectors, extinguishing agents
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Types of fire zone classification of a/c, detection system requirements, types of detection system, types of detectors, extinguishing agents and systems
5. Specific Objectives	17.2.1.1 Trainee will be able to understand the need of oxygen systems, types of systems, working principles of components.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of Fire Zone Classification Charts and Detector Types; Instructor supervision during Q&A; session.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary purpose of the aircraft fire detection system?
(b) List three common locations for fire zones in an aircraft.
(c) Why is rapid detection critical in engine compartments?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Fire zone classification types • Detection system requirements • Types of detection systems	L & Ds	PPT, IB	15 min
(b)	• Types of detectors • Extinguishing agents • Extinguishing systems	L & Ds	PPT, PDF	20 min
(c)	• System integration overview • Operational limitations • Maintenance considerations	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the difference between continuous loop and spot detection systems.
(b) Identify two key requirements for effective fire detection in cargo holds.
(c) Describe the role of Halon in modern fire extinguishing systems.

12. Home Assignment and Follow Up	Prepare a brief note on the transition from Halon to alternative extinguishing agents, referencing the next lesson on oxygen systems.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Types of oxygen systems, oxygen storage, regulators and masks	Refer to the master syllabus reference list	LP 26

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min